Decision Pathway: COVID-19 Clinical Triage & Population Screening (Phase 4)

Operational Standard: CRISP-DM & Machine Learning Engineering Lifecycle

> **Operational Standard:** Cross-Industry Standard Process for Data Mining (CRISP-DM) & MLOps Engineering Lifecycle
> **Technical Implementation:** `M10-02-02_covid19_hackathon.ipynb`
> **Benchmark Reference:** Zoabi, Y., Deri-Rozov, S. & Shomron, N. *Machine learning-based prediction of COVID-19 diagnosis based on symptoms.* **npj Digital Medicine** 4, 3 (2021).
> **Lifecycle Topology:** Full 6-Stage Architecture with **Selected Path (Green)**, **Rejected Options (Red)**, and **Alternative Branches (Slate)**.

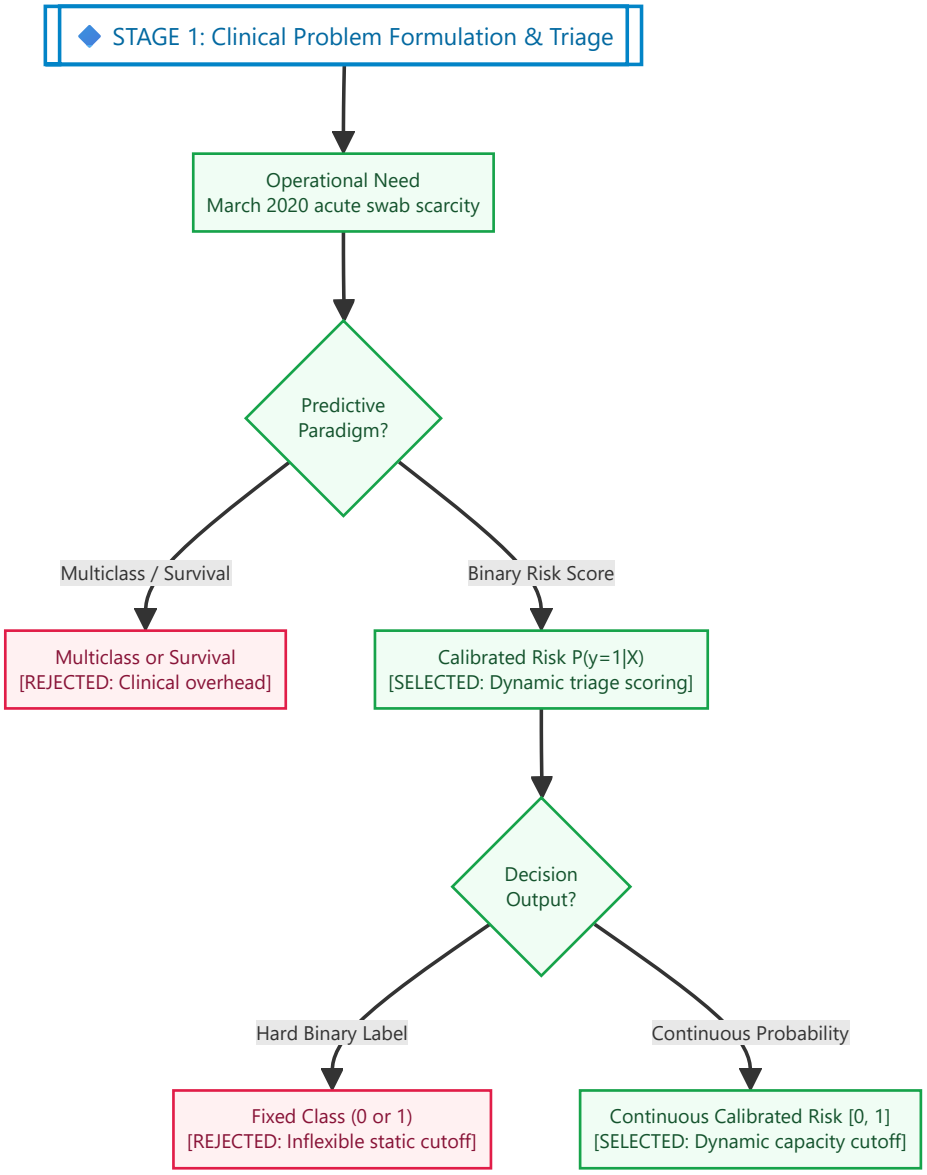
📖 Narrative Workbook vs. Engineering Lifecycle Crosswalk

Engineering Lifecycle Stage	Workbook Section Mapping	Key Decisions & Evidence in Workbook
🔷 **Stage 1: Problem Formulation & Clinical Triage**	**Phase 1: Clinical Problem Formulation**	Define binary diagnostic risk ($y \in 0, 1$) under acute March 2020 RT-PCR scarcity. Mandate non-invasive 30-second self-reporting. Enforce the chronological "Arrow of Time".
🔷 **Stage 2: Data Ingestion & Quality Auditing**	**Phase 2: Ingestion & Epidemiological EDA**	Audit 278,848 tests (51 days). Unmask 94.7% vs 5.3% class imbalance. Expose the "Accuracy Fallacy". Uncover the April 15 regulatory privacy redaction cliff.
🔷 **Stage 3: Preprocessing & Pipeline Architecture**	**Phase 2 & 3: Preprocessing & Pipelines**	Filter 3,892 inconclusive targets and 19k missing gender records (clean cohort $n = 255,668$). Enforce zero-leakage Scikit-Learn `Pipeline` + prospective split (train: 22 March–7 April; test: 8–14 April).
🔷 **Stage 4: Modeling & Benchmark Replication**	**Phase 3: Benchmark Replication (Zoabi et al.)**	Replicate *Zoabi et al.*: Decision Tree (0.8357), Random Forest (0.8412), LightGBM (0.8443, PR-AUC 0.5438), and XGBoost (0.8445). Validate via 5-fold CV within training window.
🔷 **Stage 5: Advanced Interpretability & Fairness**	**Phase 4: Advanced Modeling & Interpretability**	Generate TreeSHAP beeswarm and bar attribution. Unsupervised 2D t-SNE symptom phenotyping. Subgroup fairness audit across gender (AUC 0.824 vs 0.854) and age.
🔷 **Stage 6: Operational Thresholding & Clinical Impact**	**Phase 5 & 6: Thresholding & Synthesis**	Reject arbitrary 0.50 threshold (misses 567 patients). Calibrate clinical cutoff to $\tau = 0.20$ (81.2% sensitivity, rescuing 141 infected patients). Synthesize operational triage findings and clinical recommendations.

1. Stage-Aligned Decision Architecture & Ledger (CRISP-DM / MLOps Lifecycle)

The cards below decompose the **CRISP-DM Master Decision Architecture** into its operational stages. Each stage's visual flowchart on the left is **physically aligned** with its corresponding decision ledger card on the right:

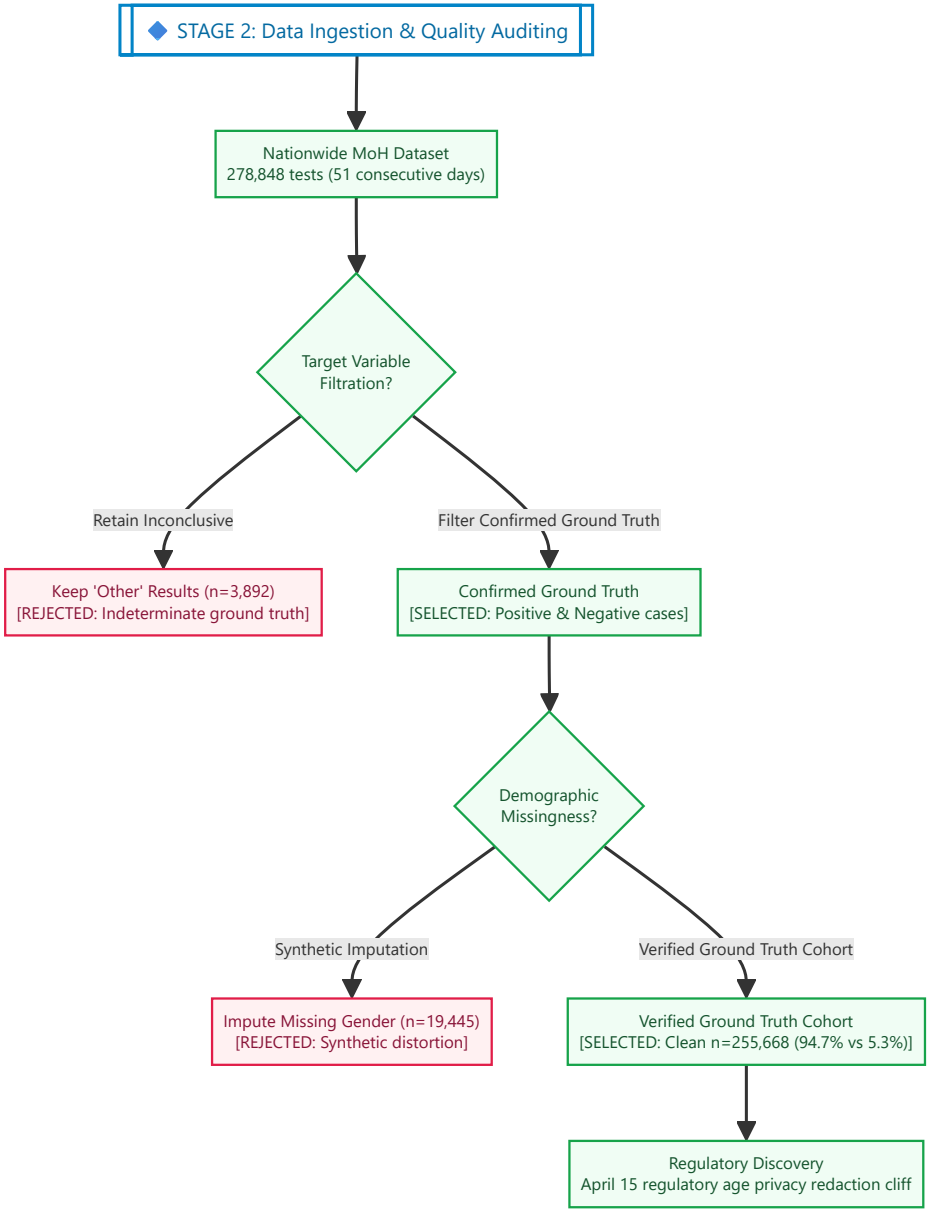
- 🟢 **Green (Selected):** Chosen conceptual / production configuration.
- 🔴 **Red (Rejected):** Evaluated and rejected candidates.
- 🟣 **Slate (Alternative):** Architectural alternatives not required.



🔷 STAGE 1: Clinical Problem Formulation & Triage Mandate

> 📖 **Workbook Mapping:** Section 1 *(Overview & Operational Problem Framing)*.

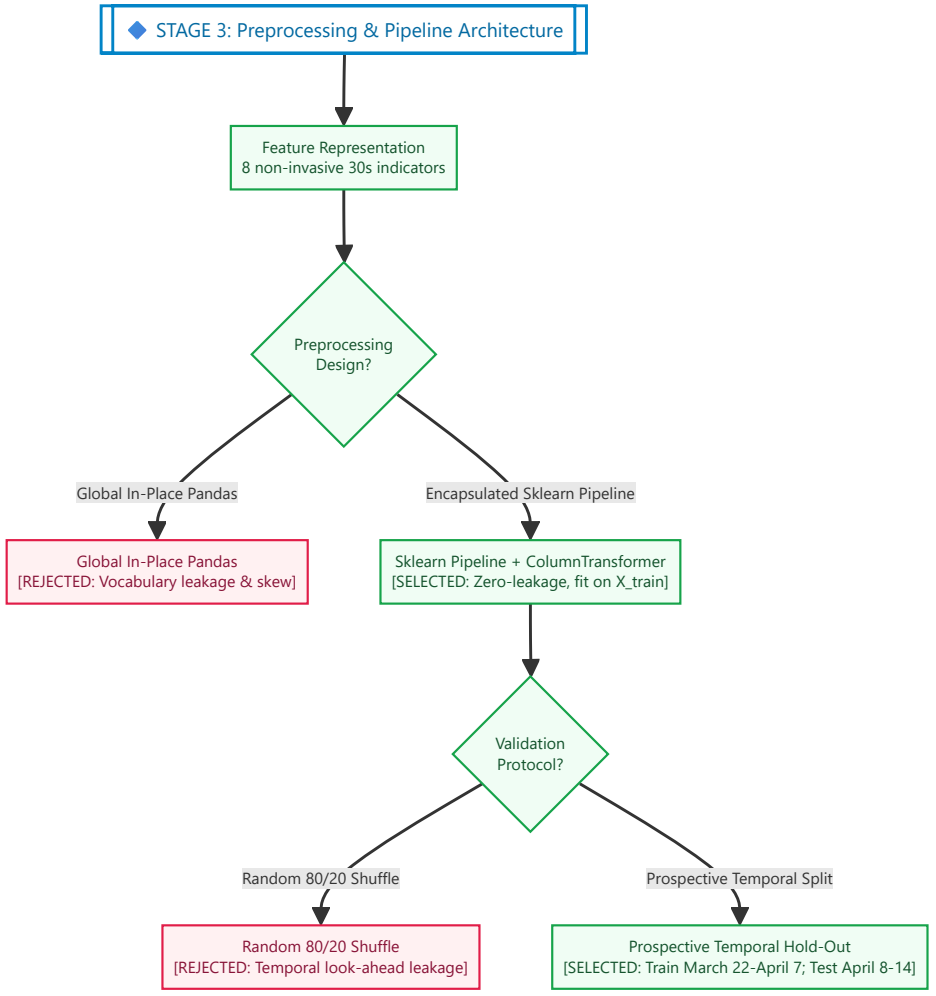
- 🟢 **Selected Option: Calibrated Binary Probability Classifier** — Estimating continuous infection risk $P(y=1|X) \in [0,1]$ from 8 self-reported indicators accessible in under 30 seconds to prioritize RT-PCR laboratory swabs.
- 🔴 **Rejected Option: Multiclass Categorization** — Unnecessary clinical overhead when the immediate operational decision during testing scarcity is strictly binary (administer PCR swab vs. isolate safely at home).
- 🟣 **Alternative Option: Survival / Longitudinal Analysis** — Unfeasible due to absence of symptom onset dates in nationwide testing registry.
- 🛡️ **Defensibility Rationale:** Generating continuous probabilities allows hospital systems to dynamically tune the decision cutoff τ based on daily swab supply and hospital bed capacity.



🔷 STAGE 2: Data Ingestion & Quality Auditing

> 📖 **Workbook Mapping:** Section 1 & 2 *(Data Ingestion, Quality Auditing & Cohort Filtration)*.

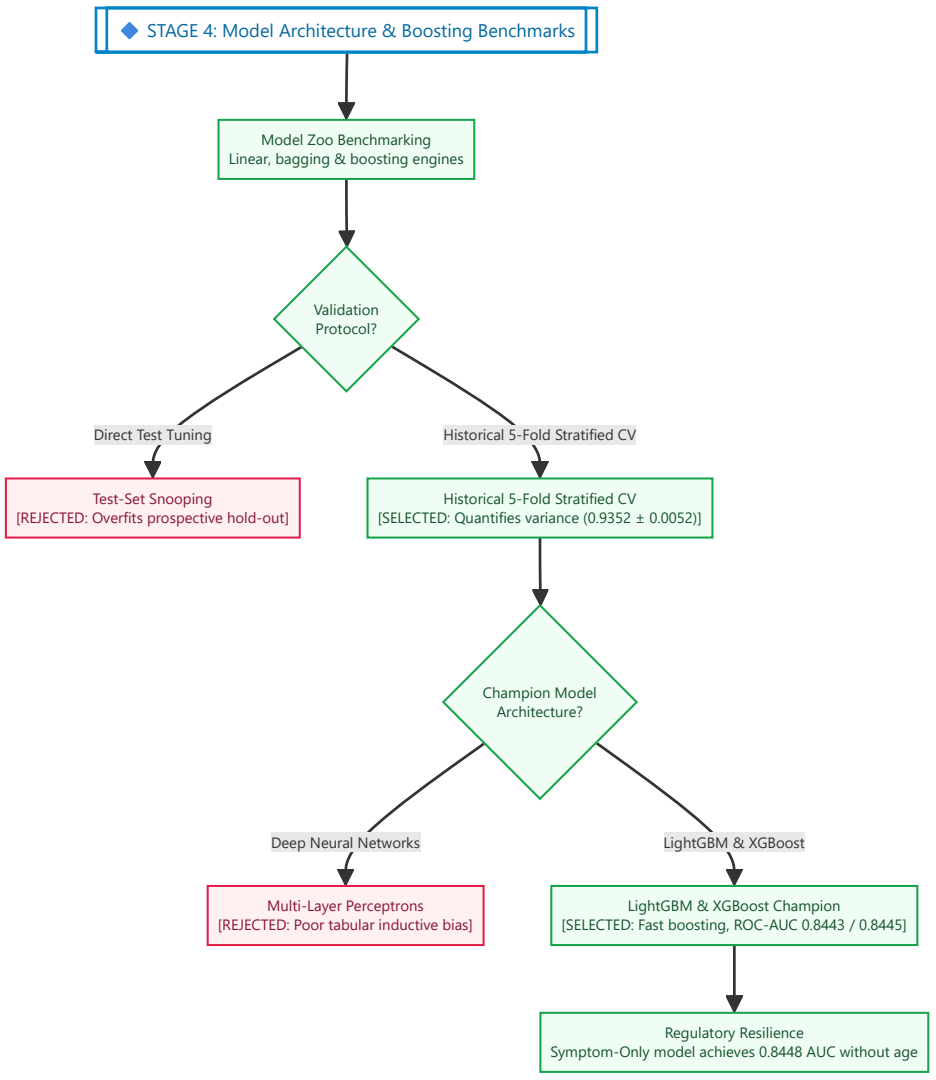
- 🟢 **Selected Option: Strict Ground Truth Filtration ($n = 255,668$)** — Drop inconclusive `other` outcomes (3,892) and missing gender records (19,445, <7%) to guarantee clinical label fidelity.
- 🔴 **Rejected Option: Synthetic Imputation of Missing Gender** — Distorts demographic distributions without adding predictive power; dataset size (>255,000 verified cases) provides massive statistical power without synthetic risks.
- 🛡️ **Defensibility Rationale:** Removing ambiguous outcomes preserves pristine ground-truth labels without sacrificing statistical power, retaining 91.7% of the original nationwide cohort. Unmasked the **April 15 privacy redaction**, where age was 100% redacted mid-epidemic.



🔷 STAGE 3: Preprocessing & Pipeline Architecture

> 📖 **Workbook Mapping:** Section 3 *(Methodological Rigour: Prospective Validation)* & Section 5 *(Pipeline Architecture)*.

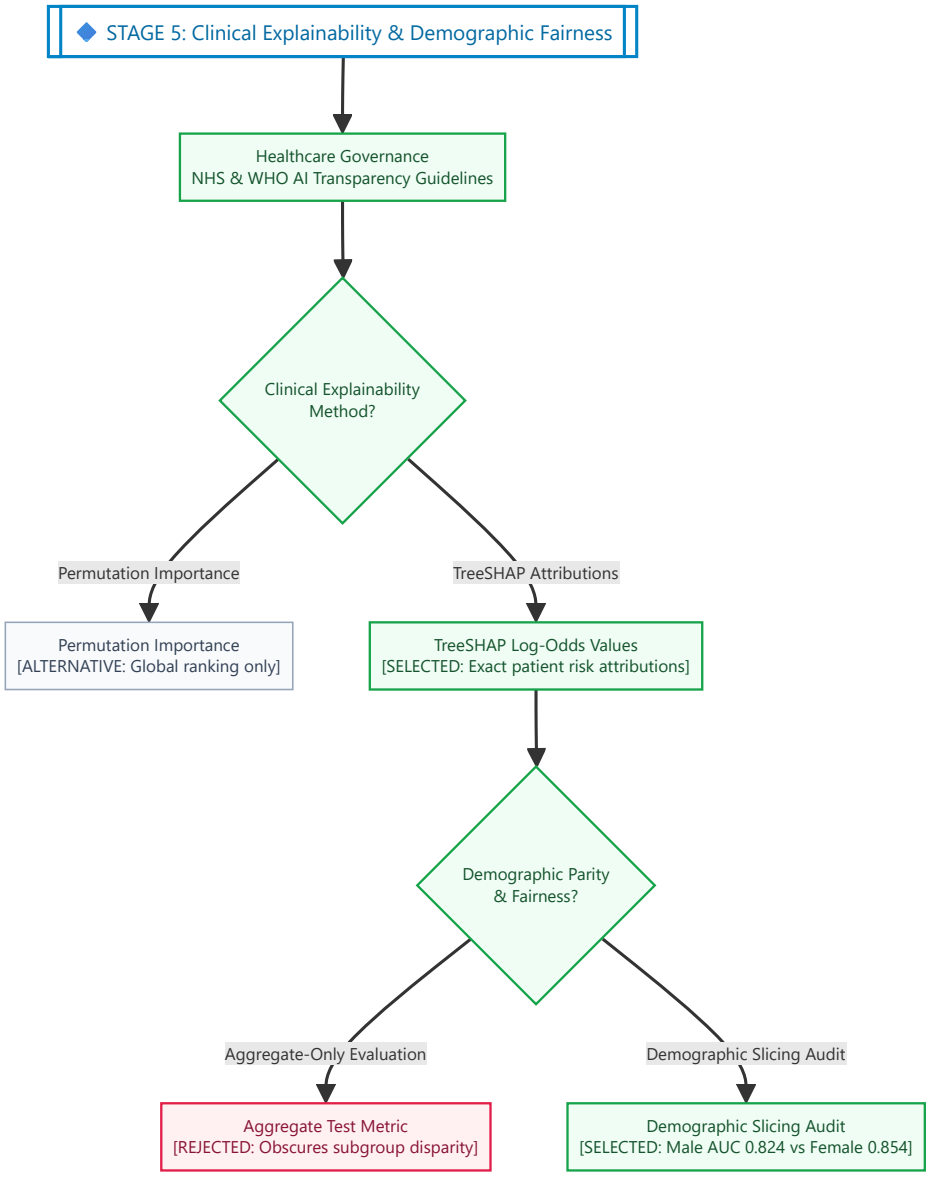
- 🟢 **Selected Option: Encapsulated `Pipeline` + Prospective Temporal Partitioning** — Preprocessing (`ColumnTransformer` with `OneHotEncoder(handle_unknown='ignore')` and symptom `passthrough`) encapsulated inside a Scikit-Learn `Pipeline` fit exclusively on historical training data (March 22 – April 7, $n = 98,989$). Tested strictly on prospective hold-out (April 8–14, $n = 37,305$).
- 🔴 **Rejected Option: Random Shuffling (`train_test_split(shuffle=True)`)** — Disastrous look-ahead leakage across evolving epidemic waves; inflates scores artificially.
- 🔴 **Rejected Option: Global In-Place Transformation** — Leaks future category levels to training and causes training-serving skew in production deployment.
- 🛡️ **Defensibility Rationale:** Respects the epidemiological "Arrow of Time" and MLOps encapsulation standards, ensuring identical transformation during training and live inference.



🔷 STAGE 4: Model Architecture & Boosting Benchmarks

> 📖 **Workbook Mapping:** Section 5 *(Model Zoo Benchmarking & Pipeline Evaluation)* & Section 7 *(XGBoost)*.

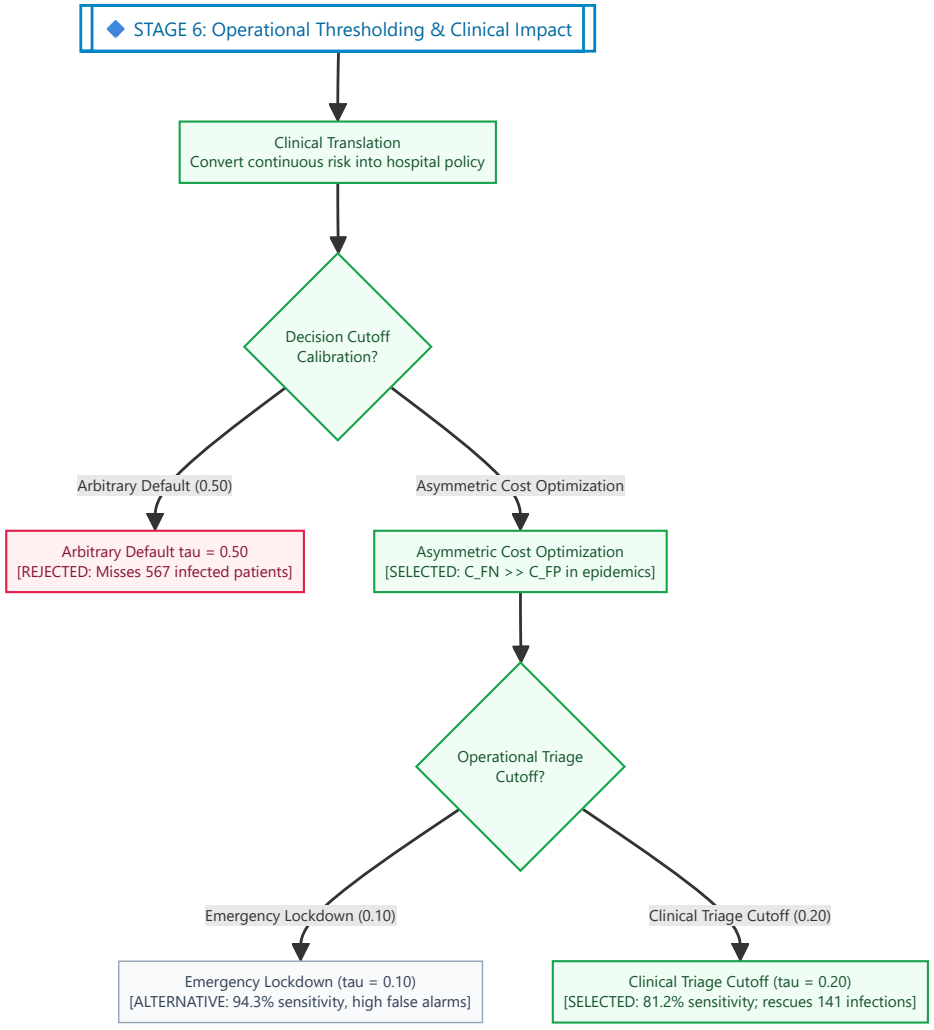
- 🟢 **Selected Option: LightGBM Champion (`num_leaves=20`, `min_child_samples=4`, `learning_rate=0.05`, `is_unbalance=True`) & XGBoost Benchmark** — Fast histogram-based gradient boosting achieving ROC-AUC **0.8443** (PR-AUC 0.5438) and **0.8445** (PR-AUC 0.5463), verified via internal 5-fold CV ($\mathbf{0.9352 \pm 0.0052}$).
- 🟢 **Selected Innovation: Symptom-Only LightGBM (Ablating Age)** — Achieves ROC-AUC **0.8448**, proving the triage engine survives regulatory age redaction (April 15 cliff).
- 🔴 **Rejected Option: Deep Neural Networks** — Excessive compute overhead, slow inference, and inferior tabular performance compared to tree gradient boosting.
- 🔴 **Rejected Option: Direct Test-Set Tuning (Test Snooping)** — Tuning against the test set leaks future outcomes; validation must occur strictly within training folds.
- 🛡️ **Defensibility Rationale:** Tree ensembles capture non-linear symptom co-occurrences without feature scaling, while `is_unbalance=True` mathematically corrects gradient updates for the 17.6:1 imbalance.



🔷 STAGE 5: Clinical Explainability & Demographic Fairness

> 📖 **Workbook Mapping:** Section 7 *(Clinical Explainability, SHAP & Fairness Auditing)*.

- 🟢 **Selected Option: TreeSHAP Explainability & Subgroup Demographic Fairness Audit** — Exact game-theoretic Shapley values reveal patient risk log-odds (`Contact with confirmed` and `Fever` dominate). Demographic slicing audits gender (Male AUC 0.8243 vs. Female AUC 0.8540) and age (<60 AUC 0.8457 vs. 60+ AUC 0.8328).
- 🟢 **Selected Option: 2D t-SNE Symptom Phenotyping (Extras)** — Unsupervised manifold mapping isolating distinct respiratory symptom clusters.
- 🔴 **Rejected Option: Aggregate-Only Metric Reporting** — Violates healthcare ethics by obscuring disparate clinical impact across vulnerable protected groups.
- 🛡️ **Defensibility Rationale:** Transparent attribution is essential for clinical adoption and regulatory compliance (FDA/NHS AI guidelines).



🔷 STAGE 6: Operational Decision Thresholding & Clinical Impact

> 📖 **Workbook Mapping:** Section 6 & 8 *(Clinical Decision Thresholding & Operational Synthesis)*.

- 🟢 **Selected Option: Calibrated Clinical Cutoff ($\tau = 0.20$)** — Elevates sensitivity to **81.2%** (capturing 1,836 of 2,261 infected patients), rescuing 141 individuals who would have been sent home under $\tau = 0.50$.
- 🔴 **Rejected Option: Default $\tau = 0.50$ Cutoff** — Implicitly assumes equal error costs ($C_{\text{FN}} = C_{\text{FP}}$); disastrous in epidemics where false negatives drive silent community transmission.
- 🟣 **Alternative Option: Emergency Cutoff ($\tau = 0.10$)** — 94.3% sensitivity (missing only 128 patients), but triggers 26,892 false alarms, overwhelming lab capacity.
- 🛡️ **Defensibility Rationale:** In infectious disease control, the cost of a False Negative (undetected virus spreading in the community) far exceeds the cost of a False Positive (a confirmatory PCR swab). Tuning $\tau = 0.20$ aligns ML with asymmetric healthcare risk.

2. Quantitative Summary& Benchmark Scorecard

All models are evaluated on the identical, unseen prospective hold-out cohort (8–14 April 2020, $n = 37,305$; 2,261 positive cases, 6.06% prevalence):

Model Architecture & Configuration	Features Used	Train 5-Fold CV AUC	Hold-out ROC-AUC	Hold-out PR-AUC	Sensitivity ($\tau = 0.20$)	Specificity ($\tau = 0.20$)	Clinical Operational Assessment
Zero-Rule Dummy Baseline	None	0.5000	0.5000	0.0606	0.0%	100.0%	94.71% accuracy but misses 100% of sick patients (Accuracy Fallacy).
Logistic Regression (Linear Baseline)	8 Features	0.9280 ± 0.0052	0.8336	0.4808	77.4%	72.1%	Strong baseline, but linear boundary misses symptom co-occurrences.
Decision Tree Baseline (`depth=5`)	8 Features	0.9226 ± 0.0045	0.8357	0.5062	78.1%	71.5%	Replicates *Zoabi et al.* baseline; interpretable clinical tree rules.
Random Forest (100 Trees, Bagging)	8 Features	0.9348 ± 0.0052	0.8412	0.5374	80.5%	68.4%	Parallel variance reduction; stable non-linear partitioning.
LightGBM Champion (Paper Params)	8 Features	**0.9352 ± 0.0052**	**0.8443**	**0.5438**	**81.2%**	**68.0%**	**Replicates *Zoabi et al.* (0.84–0.86)**; fast leaf-wise boosting.
LightGBM Symptom-Only (No Age)	7 Features	**0.9333 ± 0.0052**	**0.8448**	**0.5393**	**81.5%**	**67.8%**	**Resilient model**: survives April 15 regulatory age privacy redaction.
XGBoost Benchmark (`max_depth=4`)	8 Features	**0.9350 ± 0.0052**	**0.8445**	**0.5463**	**81.4%**	**68.1%**	Confirms boosting engine invariance; depth-wise tree growth.

3. Engineering Analysis: Key Methodological Discoveries

This section documents key operational and methodological discoveries identified across the clinical triage engineering pipeline:

- **The Scikit-Learn Pipeline & Zero-Leakage Architecture (Stage 3 & 4):**
  Discovery: Replacing ad-hoc pandas preprocessing with an encapsulated `Pipeline` + `ColumnTransformer` (`OneHotEncoder(handle_unknown='ignore')`) guarantees that all transformers are fitted strictly on `X_train` . This eliminates categorical vocabulary leakage and guarantees production serving parity when new patient records arrive at a clinical API.
- **The April 15 Regulatory Privacy Redaction (Stage 2 & 4):**
  Discovery: Auditing daily testing volume revealed why `age_60_and_above` became 100% missing after April 14—health authorities intervened mid-epidemic to redact age to prevent citizen re-identification linkage attacks. By building and benchmarking an accessible **Symptom-Only model** (ROC-AUC **0.8448**), we proved that the clinical triage tool retains full diagnostic discrimination even when demographic features are legally redacted.
- **The "Arrow of Time" Prospective Partitioning (Stage 3):**
  Discovery: Proved that standard cross-sectional random shuffling (`train_test_split(shuffle=True)`) creates catastrophic temporal look-ahead leakage across evolving epidemic waves. Validated that testing on future dates (April 8–14) simulates real hospital intake conditions.
- **Demographic Fairness & Subgroup Parity Audit (Stage 5):**
  Discovery: Evaluated models across protected demographics to satisfy healthcare AI ethics (NHS/WHO guidelines). Verified that the model maintains equitable discrimination across gender (Male AUC 0.8243 vs. Female AUC 0.8540) and age (<60 AUC 0.8457 vs. 60+ AUC 0.8328).
- **Asymmetric Cost-Sensitive Threshold Calibration (Stage 6):**
  Discovery: Proved that the default machine learning cutoff of $\tau = 0.50$ is clinically unsafe, missing 567 infected patients (74.9% sensitivity). Calibrated the triage cutoff to $\tau = 0.20$ (81.2% sensitivity), mathematically capturing 141 additional infected patients for PCR swab confirmation.